

Design Criteria Report — Final (Base Case)

Salt + Water Suspension System — 10% (w/w) salt slurry

Throughput: **100 short tons/hr (200,000 lb/hr)** dry table salt → 10% mass slurry in a **10,000 US-gal** covered stainless tank, pumped downstream.

Location: Charlotte, NC (preliminary geotechnical assumption used).

Report status: Final base-case design criteria and preliminary cost & civil/utility addendum (includes updated civil, stormwater, grading, containment, survey, geotech, instrumentation-air, and safety system costs). All costs are order-of-magnitude planning estimates in 2025 USD unless noted. A 30% contingency is applied to the grand total.

Table of contents

1. Executive summary
 2. Design basis & explicit assumptions (imperial units)
 3. Process mass balance and flows (detailed)
 4. Tank — geometry, internals, and utilities
 5. Belt conveyor — geometry, belt selection, sizing and drive
 6. Discharge chute and transfer / dust control
 7. Pump selection, hydraulic sizing, piping and NPSH considerations
 8. Agitator selection and mixing basis
 9. Instrumentation & controls (PLC architecture)
 10. Structural & civil (platform, foundations, pedestals, site footprint & earthworks) — **UPDATED**
 11. Electrical — motor loads, transformer & distribution sizing
 12. Thermal control — heat trace & insulation summary
 13. Materials of construction, safety and access
 14. Open items (data required for final design)
 15. Deliverables & schedule summary
 16. Preliminary cost estimate (itemized, with 30% contingency) — **UPDATED**
 17. Codes, standards & references
- Appendices: calculations, equipment list, preliminary P&ID sketch, instrument list, civil calculations, earthwork & grading tables
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1. Executive summary

Design for a system to accept **100 short tons/hr (200,000 lb/hr)** of table salt delivered on an inclined troughed belt conveyor and produce a **10% (w/w)** salt-in-water slurry in a **10,000 US gallon** covered 304-SS tank elevated **5 ft** above final grade. Slurry pumped at design TDH = **100 ft**, base-case flow $\approx$ **3,733 GPM**. Prime mover selections: pump motor **175 HP VFD**, conveyor motor **15 HP fixed-speed**, agitator **15 HP fixed-speed**. Controls: local PLC + HMI with field instrumentation (level, temperature, flow, pressure, belt scale, safety interlocks). This final report incorporates a detailed civil & utilities addendum: earthwork cut/fill estimates, grading & pavement, stormwater conceptual design (1-inch capture basis), culvert and swale recommendations, secondary containment sized to 110% of tank volume, survey & geotechnical budgets, and an instrument-air compressor package. Updated cost tables reflect

these additions; grand total (with 30% contingency) $\approx$ **\$1,147,380** (includes optional fire pump allowance; remove optional fire pump to reduce cost by $\sim$ \$50k).

2. Design basis & explicit assumptions (all imperial units)

(As already developed; key items summarized and unchanged except for additions noted)

- Feed: **100 st/hr = 200,000 lb/hr** dry NaCl.
 - Particle size: **100–500 μm** . Salt moisture $\leq 0.5\%$.
 - NaCl true density: **135.6 lb/ft³**. Bulk density (loose): **$\sim$ 74.91 lb/ft³**.
 - Slurry concentration (design): **10% mass**. Process temperature 60 °F. 10% NaCl solution density **$\sim$ 8.93 lb/gal** (SG $\approx$ 1.07).
 - Conveyor: 24 in belt (2 ft), incline 10°, horizontal projection 100 ft, nominal belt speed 197 ft/min.
 - Tank: 10,000 US gal, 304 SS, covered, elevated on steel platform 5 ft above grade, H:D = 2:1.
 - Pump: centrifugal slurry pump, 316 SS wetted parts, pump efficiency assumed 70%, motor efficiency 95%, preliminary motor **175 HP VFD**. Suction piping recommended 24 in, discharge 20 in (confirm with vendor).
 - Geotech/Foundations initial allowable bearing = **3,000 psf** (Charlotte assumption). Final geotechnical required.
 - Existing site slope assumed **0.25%** (0.0025) for initial grading assumptions (per user note).
 - Codes: ASME B31.3, ACI 318, AISC, NFPA, NEC, OSHA, API 610, etc.
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3. Process mass balance and hydraulic flows (detailed)

(Full calculations retained from earlier report — summary numbers below.)

- Salt feed: **200,000 lb/hr**.
- Water required for 10% mass slurry: **1,800,000 lb/hr**.
- Total slurry mass: **2,000,000 lb/hr**.
- Slurry density: **8.9296 lb/gal** (10% NaCl @ 60 °F).
- Volumetric flow: **$\approx$ 223,975 gal/hr $\approx$ 3,733 GPM**.
- Pump hydraulic HP @ 100 ft TDH: **$\sim$ 100.9 HP**; motor HP (70% pump eff, 95% motor eff) $\approx$ **151.7 HP** $\rightarrow$ select **175 HP VFD**.

(See Appendix A for step-by-step math.)

4. Tank — geometry, internals, and utilities

- Volume = 10,000 gal = 1,336.81 ft³. H:D = 2:1 → **D ≈ 9.48 ft, H ≈ 18.95 ft**.
 - Internals: 4 baffles, top-entry agitator (15 HP), 36" manway, sample port, vent with flame arrestor if required, drain 3" valve. Insulation & heat trace to maintain 60 °F.
 - Loads (loaded): slurry mass = 89,296 lb; tank structure & internals estimated +10,000 lb → **total ≈ 99,296 lb** distributed to supports. Design supports per ACI & AISC. Foundation pad sizing prelim: 3 ft × 3 ft × 2 ft pads (4 pads).
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5. Belt conveyor — geometry, belt selection, sizing and drive

- 24 in belt (2 ft), length incl. allowances ≈ **101.5 ft** run, vertical rise ≈ **17.63 ft**, nominal belt speed 197 ft/min.
 - Salt volumetric feed ≈ **2,669.74 ft³/hr** ; loaded depth on 2 ft belt ≈ **1.355 in**.
 - Conveyor motor recommended **15 HP** (fixed speed). Include guards, pull cords, misalignment switches, E-stops, belts with abrasion cover, transfer skirt and chute lining.
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6. Discharge chute and transfer / dust control

- Lined skirted chute with impact plate and hood; transfer to tank top with sealed feed-in to minimize fugitive dust. Provision and space for small baghouse/local exhaust if dust control required (OPEN ITEM).
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7. Pump selection, hydraulic sizing, piping and NPSH considerations

- Design flow **3,733 GPM**, TDH **100 ft**, slurry SG **1.07**. Hydraulic HP ~100.86 HP. Motor selected **175 HP VFD** as planning size.
 - Pump: centrifugal slurry pump, 316 SS wetted parts; heavy-duty shaft and bearings. Consider lined casing or hardfacing if solids cause abrasion. Provide suction screening, straight suction run per vendor, and suction piping at least equal or larger (recommend **24 in suction**). Discharge piping **20 in** (confirm). Magnetic flowmeter for discharge. Compute NPSHa once final pump location and tank/pump elevations are available.
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